IoT Botnet Attack Detection

- Jinhong Lin
- Elton Chang
- Shiwei Jiang

Brief Description (Proposal Paragraph)

As Internet of Things (IoT) devices proliferate in homes, offices, and critical infrastructure, they have become prime targets for malware and botnet attacks. The Mirai and Gafgyt botnets have demonstrated the devastating potential of compromised IoT devices—turning ordinary cameras, routers, and smart home devices into powerful attack vectors for distributed denial-of-service (DDoS) attacks.

The challenge lies in accurately distinguishing between benign IoT traffic and malicious botnet activity in real-time, enabling network administrators to rapidly detect and isolate compromised devices before they can be weaponized. This dataset provides a robust, labeled benchmark with over 9.3 million network flow samples capturing authentic IoT device behavior alongside multiple botnet attack variants. The high-quality engineered features extracted from 10-second traffic windows enable machine learning models to learn discriminative patterns between attack and normal behavior, making it an ideal platform for developing and validating next-generation IoT intrusion detection systems.

Our group project aims to answer the following research questions. How can we accurately detect and classify IoT botnet attacks (Mirai and Gafgyt) in real-time network traffic using machine learning models trained on network flow statistics?

And the ultimate goal is proactive defense—catching compromised devices before they can participate in large-scale attacks like the 2016 Mirai DDoS that took down major websites.

We will be using the BoTNeTIoT-L01 IoT intrusion detection dataset from Kaggle to study IoT dataset for Intrusion Detection Systems (IDS). The dataset is large and suitable for machine learning/deep learning experiments: the cleaned version (BotNeTIoT-L01_label_NoDuplicates.csv) has about 2.43 million rows and 25 columns (~463 MB), and the extended version (BoTNeTIoT-L01-v2.csv) has about 7.06 million rows and 27 columns (~2.7 GB) with metadata such as device name, attack family (Mirai/Gafgyt), and attack subtype.